Phase-Wave Differentials: A Receiver-Relative Signal Operator for Testing Neural State Updates

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Abstract

A neural signal can differ from its recent context in several ways at once: relative phase, amplitude, duration, and spatial propagation. Existing analyses often measure these quantities separately or summarize interaction with a scalar synchrony statistic. This paper defines a **phase-wave differential** (PWD) as a receiver-relative, edge-conditioned vector of departures from a strictly past baseline. The construction starts with a causal complex filter bank, expresses phase at a receiver relative to a putative sender, estimates circular-phase and log-amplitude references using only earlier samples, and combines wrapped phase departure, log-amplitude departure, event duration, and a coordinate-normalized lagged phase gradient. The operator is invariant to a common phase rotation and, for its amplitude-departure component, to common positive amplitude scaling. Its interpretation is conditional on the selected filter, reference, receiver, edge direction, delay, phase-quality gate, and spatial coordinates. Delay and expected phase offset are not generally identifiable from one narrow band; low-amplitude phase, common input, volume conduction, waveform asymmetry, evoked latency fields, and reference choice can all mislead.

An existing implementation passed eight deterministic causal and shape checks on generated signals, while a separate equal-input synthetic comparator did not show a PWD-feature advantage. A frozen Allen Visual Coding Neuropixels development procedure then compared an identical 16-feature shared raw branch plus eight PWD random features against the same shared branch plus eight additional raw random features. Fourteen development sessions, 921 eligible VISp units, three fixed projection seeds, and 908,295 causal samples were analyzed. Every session-level primary increment was negative: the cross-session median raw-minus-PWD held-out Poisson-deviance difference was -0.001099 (descriptive session-bootstrap 95% interval -0.001301 to -0.000872; two-sided exact sign test, 0 of 14 positive, p=0.000122). Thus, the full tested PWD branch underperformed the capacity-matched raw-only branch.

The controls narrow that failure. Intact PWD outperformed its within-block circular-shift null and the source-receiver role swap in all 14 sessions. Removing amplitude and duration also worsened the PWD branch in all 14 sessions. By contrast, removing the phase components improved performance in all 14 sessions. These controls do not rescue the failed primary benchmark; they indicate that the tested phase representation was the most immediate redesign target, while aligned, directional, amplitude-duration structure retained limited predictive information relative to its own nulls. The five-session final test remains unopened. PWD is therefore retained as a defined candidate observable, while the present 25-feature implementation fails its grouped development criterion. The original protocol terminates at that adverse result. Any redesigned model must be treated as a new, prospectively frozen successor rather than a continuation or confirmation of the failed model.

signal processing; circular statistics; neural oscillations; identifiability

Keywords: neural phase; phase of firing; traveling waves; receiver-relative coding; causal

Scope and evidence boundary

a frozen-procedure analysis of 14 coordinate-eligible sessions from a previously published mouse Neuropixels dataset. The primary development comparison was unfavorable to the tested PWD feature set in every session. The original protocol is closed at this adverse development endpoint without spending its five sealed final-test sessions. Those sessions remain unopened and may be used only after a separately named successor model and analysis protocol have been learned exclusively from development data, independently reviewed, and frozen. This result does not establish that PWDs are used by brains, reconstruct conscious content, or prove SAN or NAPOT.

1. The problem before the acronym

Suppose two neural populations are active at the same average rate and power. Their effects on a third population may still differ because their events arrive at different moments in that population's excitability cycle, persist for different durations, or participate in different spatial patterns. A useful analysis therefore needs more than a label such as "gamma" and more than a global synchrony score. It needs to state: difference from what baseline, measured at which receiver, on which edge, at which delay, over which band, and with what observable consequence.

The PWD proposal answers that measurement question. In plain language, a PWD is a structured record of how an incoming or locally measured pattern differs from a receiver's recent operating context. The acronym is earned only after the baseline and receiver are declared. The word *phase* identifies one circular component, *wave* allows spatial and lagged structure to be tested, and *differential* means a departure from a reference rather than an absolute signal value.

This definition is narrower than the historical SAN use of PWD as part of Neural Rendering. It does not assume that every difference carries content. It does not identify an internal screen or viewer. The scientific question is whether the measured differential predicts or causally changes an independently defined neural, perceptual, or behavioral endpoint after stronger alternatives are controlled.

2. Scope and claim classes

Four claim classes must remain separate.

1. **Established components.** Neural signals have rate, amplitude, phase, temporal, and spatial structure. Phase-of-firing codes and traveling cortical waves have empirical precedents [3-6].
2. **Defined operator.** Equations (1)-(11) specify one causal PWD estimator. The implementation and synthetic checks make this operator reproducible.
3. **Bounded empirical result.** The tested 25-feature branch did not add held-out predictive information beyond its capacity-matched raw-only comparator in any of 14 development sessions. Whether a redesigned PWD family can add information or causal leverage remains prospective.
4. **Broader SAN interpretation.** PWDs may participate in distributed neural state updating and Neural Rendering [1,2]. The operator alone cannot establish that interpretation.

The contribution of this paper is Classes 2 and the test contract for Class 3. It uses Class 1 as biological motivation while leaving Class 4 conditional.

3. Historical boundary

The developmental genealogy distinguishes functional ancestors from later terminology. In the April 2017 *Neural Lace Podcast*, Blumberg asked whether the brain's network protocol was analogous to connection-and-feedback transmission (TCP) or connectionless transmission (UDP), then proposed an **Alternating Transmission Protocol** (ATP) that could switch between those roles [20]. The name was also a deliberate reference to adenosine triphosphate, the cellular energy currency, linking the communication question to biological energy use. A related 2017 source described calculating and transmitting the difference between a present neural pattern and a desired pattern. These are functional ancestors of the later program, not the complete PWD or NAPOT mechanisms. The phase-specific formulation and the name *phase-wave differential* were recorded in the 2022 SAN corpus [1,2]. A separate July 5, 2022 note already described successive action-potential phase intervals as changing vectors and Taylor-series-like polynomial terms whose inhibitory interactions refine an approximated curve or represented pattern [19]. The file was created in commit `560df7f` and renamed to `a0258z.md` in commit `71bbb1a`; the rename must not be mistaken for the originating content change. That Taylor analogy is historical genealogy, not a derivation of Equations (1)-(20). The companion manuscript *Phase-Wave Differential Calculus: Neural Tuning, Variability Tokens, and Taylor-Sequence Approximation in Oscillatory Neural Networks* owns the broader typed-event, Taylor / Volterra/CV, and synthetic-falsification program; this paper owns the minimal causal estimator and frozen biological benchmark. The earlier operation should not be erased, and the later name should not be backdated. Chronology can establish how a research program developed; it cannot substitute for empirical validation.

4. Causal signal model

4.1 Operator signature, initialization, and declared choices

The complete estimator is a map

$$\mathcal{F}_{\Theta,t} : (\mathbf{X}_{1:t}, \mathcal{E}, \mathbf{R}) \longmapsto \{\mathbf{d}_{eft} : e \in \mathcal{E}, f \in \mathcal{B}\}\,, \tag{1}$$

where $\mathcal{E}$ is the declared edge set, $\mathbf{R}$ contains channel coordinates, $\mathcal{B}$ is the frequency set, and Θ contains filters, retention constants, thresholds, lags, numerical floors, phase-quality rules, and warm-up length. A PWD result is not reproducible if any of these choices is omitted.

Initialization is part of the estimator. In the current software fixture, the circular state starts at the positive real unit and the amplitude state starts from the first log-amplitude observation. Those conventions affect early samples. Confirmatory analysis must either discard a preregistered warm-up interval long enough for initialization influence to fall below a stated tolerance, or propagate initialization as model uncertainty. A test split cannot be used to choose the warm-up.

The edge set is also not innocent preprocessing. If candidate edges are selected from target labels or full-dataset coupling, leakage has already occurred before model fitting. Edge discovery must be external, training-only, or nested inside each training fold. Direction must remain separate from edge existence.

4.2 Observed channels and causal complex features

Let the observed multichannel signal be

$$\mathbf{x}_t = (x_{1t}, \ldots, x_{Ct})^\top, \qquad t = 1, \ldots, T. \tag{1}$$

For channel c and preregistered center frequency f, define a one-sided complex feature

$$z_{cft} = \sum_{k=0}^{K_f} h_f(k) x_{c,t-k}. \tag{2}$$

All nonzero filter support lies at the current or earlier samples. The implementation accompanying the NAPOT methods program uses a causal complex demodulator with an exponentially retained state; Equation (2) states the more general contract. Write

$$z_{cft} = A_{cft} e^{\mathrm{i}\phi_{cft}}, \qquad A_{cft} = |z_{cft}|, \qquad \phi_{cft} = \mathrm{Arg}(z_{cft}). \tag{3}$$

This phase is an estimator derived from a selected representation. It is not evidence that the raw signal is a stationary or sinusoidal oscillator. Phase is unreliable near zero amplitude and can be altered by bandwidth, waveform shape, filter edge behavior, and reference choice [9,10].

4.3 Receiver-relative edge phase

Let $e = (s \to r)$ denote a directed candidate edge from sender s to receiver r. Direction must be supplied by anatomy, intervention, or a separately validated effective- connectivity analysis. It cannot be inferred from phase difference alone. The instantaneous receiver-relative edge phase is

$$\delta_{erft} = \mathrm{wrap}(\phi_{rft} - \phi_{sft}), \quad \mathrm{wrap}(u) = \mathrm{Arg}(e^{\mathrm{i}u}). \tag{4}$$

The receiver index is retained because the same arriving pattern can have different consequences at receivers with different excitability, morphology, inhibition, or recent history.

4.4 Strictly past reference state

Let $0 < \rho_\phi < 1$ and $0 < \rho_A < 1$ be fixed retention parameters. A circular phase state is updated after the baseline for time t has been emitted:

$$\begin{aligned} q_{eft} &= \rho_\phi q_{ef,t-1} + (1-\rho_\phi) e^{\mathrm{i}\delta_{erft}}, \\ \bar{\delta}^{\mathrm{past}}_{eft} &= \mathrm{Arg}(q_{ef,t-1}). \end{aligned} \tag{5}$$

Likewise, the receiver log-amplitude reference is

$$\begin{aligned} m_{rft} &= \rho_A m_{rf,t-1} + (1-\rho_A)\log(A_{rft} + \epsilon), \\ \bar{a}^{\mathrm{past}}_{rft} &= m_{rf,t-1}, \end{aligned} \tag{6}$$

where $\epsilon > 0$ is a declared numerical floor. The `past` superscript is substantive: the sample being evaluated cannot define its own expectation.

4.5 Differential components

The wrapped phase departure is

$$p_{eft} = \mathrm{wrap}\left(\delta_{erft} - \bar{\delta}^{\mathrm{past}}_{eft}\right). \tag{7}$$

The log-amplitude departure is

$$a_{eft} = \log(A_{rft} + \epsilon) - \bar{a}^{\mathrm{past}}_{rft}. \tag{8}$$

For a threshold $\eta_f > 0$, event duration is a causal run length:

$$\ell_{eft} = \begin{cases} \ell_{ef,t-1} + 1, & |p_{eft}| \geq \eta_f, \\ 0, & |p_{eft}| < \eta_f. \end{cases} \tag{9}$$

Let $\mathbf{r}_c$ be the coordinate assigned to channel c, and let $L \geq 1$ be a fixed lag. The implemented propagation feature is

$$g_{eft} = \frac{\operatorname{wrap}(\phi_{rf,t} - \phi_{sf,t-L})}{\|\mathbf{r}_r - \mathbf{r}_s\|_2}. \tag{10}$$

This is a coordinate-normalized lagged phase gradient. It is not automatically a physical velocity, an axonal transmission speed, or a causal-flow estimate.

The operational PWD vector is

$$\mathbf{d}_{eft} = [p_{eft},\ a_{eft},\ \ell_{eft},\ g_{eft}]^\top. \tag{11}$$

Other versions may add frequency departure, phase-response-curve occupancy, curvature, or cross-scale terms. They are different estimators and must be named as such. A study cannot change the vector after inspecting test outcomes and continue to call it the same confirmatory PWD.

5. Mathematical properties

5.1 Common phase-rotation invariance

If every channel phase is rotated by the same α_{ft}, receiver-relative edge phase is unchanged:

$$\operatorname{wrap}[(\phi_{rft} + \alpha_{ft}) - (\phi_{sft} + \alpha_{ft})] = \delta_{erft}. \tag{12}$$

This removes a common angular origin at one frequency and time. It does not remove mixing from a common physical source or make the estimate reference-free.

5.2 Common amplitude-scale invariance

For a common positive constant b, scaling the receiver feature and its historical samples by b adds $\log b$ to both current and baseline log amplitude, giving

$$a'_{eft} = [\log b + \log(A_{rft} + \epsilon/b)] - [\log b + \bar{a}^{\text{past}}_{rft}] \approx a_{eft}, \tag{13}$$

with exact invariance when $\epsilon = 0$ and amplitudes are positive. With a numerical floor, the approximation must be checked near the gate. Channel-specific gains and time-varying gains do not cancel automatically.

5.3 Causality

Let F_t denote the complete operator through time t. Because Equations (2), (5), (6), (9), and (10) use no sample after t,

$$\mathbf{x}_{1:t} = \mathbf{x}'_{1:t} \Longrightarrow F_t(\mathbf{x}) = F_t(\mathbf{x}'). \tag{14}$$

The synthetic validator tests this property by replacing all future inputs with large random perturbations and requiring zero change in every present and past output.

5.4 Phase-quality gate

PWD phase components are eligible only where a preregistered quality rule is satisfied, for example

$$Q_{cft} = \mathbb{1}[A_{cft} > \epsilon_A]\mathbb{1}[N_{cft}^{\text{valid cycles}} \geq n_{\min}]\mathbb{1}[\kappa_{cft} \geq \kappa_{\min}], \tag{15}$$

where the exact amplitude, cycle-count, and circular-concentration requirements are fixed without test labels. Equation (15) is a template, not a universal biological threshold.

5.5 Properties that are and are not proved

The causal implementation establishes three bounded propositions for the configured discrete-time operator.

Proposition 1: prefix invariance. If two inputs share the same samples through t, every operator output through t is identical. This follows directly from the one-sided filter, strictly past state update, causal duration recurrence, and lagged feature. The validator tests the property under a large future perturbation.

Proposition 2: edge-phase antisymmetry. For the instantaneous edge-phase component only, reversing $s \to r$ gives the wrapped negative of the original phase. This proposition does not extend automatically to amplitude departure, duration, or the lagged spatial feature, because those components are receiver-specific or time-asymmetric.

Proposition 3: coordinate-unit covariance. If coordinates are multiplied by $c > 0$, the lagged-gradient component is divided by c. It is therefore normalized by the supplied distance but not invariant to coordinate units. Reports must state whether coordinates are sensor-space, cortical-surface, Euclidean source-space, or path-length estimates.

None of these propositions proves biological validity, optimality, unique decomposition, or incremental prediction. They are properties of the estimator.

5.6 Equation-to-code crosswalk

Mathematical object	Current implementation	Verified behavior	Open scientific issue
Equations (2)-(3), causal complex feature	causal_complex_filter_bank	shape, finiteness, and future-prefix invariance	biological band and filter choice
Equations (5)-(6), past baselines	strictly_past_baselines	current sample excluded from both references	initialization, stationarity, and context definition
Equations (7)-(10), differential components	phase_wave_differentials	wrapping, phase antisymmetry, causal duration, lag warm-up	receiver physiology, edge direction, source mixing, and geometry

Mathematical object	Current implementation	Verified behavior	Open scientific issue
Equation (11), PWD output	causal_pwd_pipeline	deterministic joined output	incremental information and causal utility

The code uses edge_phase\=\angle(receiver*\conjugate(sender)), consistent with Equation (4). Its lagged-gradient term compares current receiver phase with lagged sender phase and divides by the declared edge distance. The code comment and this manuscript both prohibit interpreting that value as physical velocity without a separate wave model.

6. Information, surprise, and meaning

The historical SAN canvas-and-ink description can be translated cautiously into information- theoretic language. Under a fitted predictive model, a stable tonic state is relatively expected and therefore often has lower sample-wise surprisal; a rare phasic departure may have higher surprisal. For an observed differential $\mathbf{d}$ under context H, define

$$S(\mathbf{d} \mid H) = -\log p(\mathbf{d} \mid H). \tag{16}$$

This does not mean that low frequency is intrinsically low information or high frequency is intrinsically high information. A predictable high-frequency train can have low surprisal, and a slow unexpected transition can have high surprisal. Nor does high surprisal establish semantic content. The relevant empirical quantity is incremental information about an independently defined target Y:

$$I(Y; \mathbf{D} \mid \mathbf{B}) = H(Y \mid \mathbf{B}) - H(Y \mid \mathbf{B}, \mathbf{D}), \tag{17}$$

where $\mathbf{B}$ contains rate, power, scalar synchrony, connectivity, raw history, nuisance variables, and any other eligible baseline information. Shannon's formalism [12] supports this measurement distinction; it does not by itself establish Neural Rendering, consciousness, or the physical interpretation proposed by Super Information Theory.

A second distinction is necessary. Sample-wise surprisal is computed under a probability model, whereas the conditional information in Equation (17) is a population quantity. In practice, the primary claim should be evaluated with nested out-of-sample log loss or another proper scoring rule, not a post hoc plug-in mutual-information estimate selected because it is favorable. Any information estimate must declare its estimator, bias correction, uncertainty procedure, and effective sample unit.

7. Identifiability and failure conditions

7.1 Delay-offset equivalence

In a narrow band with angular frequency ω_f, an expected offset ψ_{ef} and delay τ_e enter approximately as

$$\delta_{ef}(t; \tau_e, \psi_{ef}) \simeq \mathrm{wrap}[\phi_r(t) - \phi_s(t) + \omega_f \tau_e - \psi_{ef}]. \tag{18}$$

Multiple (τ_e, ψ_{ef}) pairs therefore produce the same wrapped relation. Identification requires independent delay information, a constrained multifrequency model, intervention, or an explicit equivalence class with propagated uncertainty.

7.2 Measurement non-uniqueness

The following are constitutive failure risks:

- **Low amplitude:** phase becomes unstable when the analytic vector approaches the origin.
- **Filter dependence:** center frequency, bandwidth, order, and edge handling change phase.
- **Waveform shape:** nonsinusoidal activity can produce harmonics that resemble separate bands.
- **Reference and mixing:** one source can appear at multiple sensors; zero-lag coupling can be inflated by volume conduction or field spread [7,8].
- **Common input:** two channels can share phase without direct transfer.
- **Evoked latency fields:** orderly response delays can resemble a propagating wave.
- **Geometry error:** sensor coordinates may not equal source coordinates, and Euclidean distance may not represent cortical or axonal path length.
- **Direction error:** an ordered software edge is not biological direction evidence.
- **Receiver mismatch:** recorded LFP phase is not automatically a receiver's excitability phase.
- **Multiple comparisons:** searching bands, lags, edges, thresholds, and endpoints can manufacture an apparent optimum.

Lag-resistant statistics such as the imaginary part of coherency, phase-lag index, or weighted phase-lag index can reduce specific zero-lag mixing problems [7,8]. Phase-locking statistics also require an explicit estimator and sampling model [11]. These methods do not solve source mixing, direction, or biological interpretation in general.

8. Existing software-contract result

The current causal implementation is stored in the NAPOT methods package and is bound by SHA-256 in this paper's Draft 2 source manifest. The causal validator was freshly replayed on 2026-07-19 and passed 8/8 checks. The separate SAN suite was also replayed in its pinned CPython 3.12.10 and NumPy 2.4.6 environment: all 19 tests passed normally and all 19 passed with Python optimization. On a generated fixture with 2,400 samples, three channels, two directed edges, and 9, 28, and 60 Hz center frequencies, all eight declared checks passed:

1. wrapped phase remained in the half-open interval $[-\pi, \pi)$;
2. reversing an edge reversed receiver-relative phase to numerical precision;
3. the phase baseline at time t excluded sample t;
4. the amplitude baseline at time t excluded sample t;
5. the duration counter was causal and reset correctly;
6. output shapes matched the contract;
7. all outputs were finite after lag warm-up; and
8. changing all future inputs changed no present or past output, with maximum difference exactly zero.

These are software mechanics, not neuroscience evidence. The configured time constants, threshold, lag, bands, and geometry are fixture inputs. The code has not established a biological receiver baseline, source direction, neural mechanism, target-variable increment, or replication.

A separate SAN synthetic benchmark gave a flexible raw-input model and a PWD / NAPOT model the same raw access, folds, nonlinear family, width, trainable parameter count, and fold-level seed. The stronger comparator removed the earlier favorable constructed direction; the recorded contrast was M5-M6\=\-0.01 14485499. Nineteen software tests passed, but this unfavorable synthetic comparison provides no evidence that hand-engineered PWD features outperform an equal-input raw model. It is included because a negative comparator result is part of the evidential record, not a detail to be hidden.

9. Biological component evidence

Voloh, Oemisch, and Womelsdorf recorded fronto-striatal activity in nonhuman primates and reported that firing at different beta phases carried different information about outcome, reward-prediction error, and outcome history, including phase-dependent information beyond a phase-blind rate code [3]. This supports the component proposition that phase relative to an ongoing population signal can matter for encoded variables. It does not validate the PWD vector, its baseline, its spatial term, or the SAN interpretation.

Huxter, Burgess, and O'Keefe found dissociable rate and theta-phase relations in hippocampal place cells [4]. Muller and colleagues reported stimulus-evoked propagating population activity in awake monkey visual cortex [5]. Aggarwal and colleagues reported fast feedforward and slower feedback traveling waves in awake mouse visual cortex, with cross-frequency interaction and phase locking of spiking [6]. Together these studies make phase, relative timing, and propagation legitimate observables. They do not show that the joined PWD operator is the brain's code or that it reconstructs experience.

9.1 Receiver phase and routing are experimentally separable components

Llinas's electrophysiological synthesis distinguishes neurons that act as intrinsic pacemakers from neurons that respond preferentially to selected input frequencies because of their voltage-dependent conductances [15]. This strengthens the reason to retain the receiver index: a signal's consequence depends on the receiving cell's dynamical state. It does not establish PWD, a microtubule source, or consciousness. Cantero et al. reported a prominent 39 Hz electrical oscillation in voltage-clamped brain-microtubule bundles and oscillatory conduction in membrane-permeabilized neurites [16]. That preparation-level result motivates an exploratory 30-55 Hz sensitivity analysis, not a claim that microtubules generate intact cortical gamma.

Cardin and colleagues used cell-type-targeted optogenetic stimulation in mouse barrel cortex and reported that fast-spiking interneuron activation induced gamma-range network states; the timing of a sensory input relative to the gamma cycle changed the amplitude and precision of the evoked response [13]. This is direct component evidence for a phase-dependent receiver window. It does not show that the four-component PWD vector is the causal variable, and it does not generalize one gamma preparation into a universal band code.

Colgin and colleagues reported that fast and slow gamma regimes in hippocampal CA1 were associated with different coupling to medial entorhinal cortex and CA3, respectively, and occupied different theta phases [14]. This supports preparation-specific frequency / phase routing as an empirical phenomenon. It does not establish that frequency alone determines source, that the relationship is universal, or that a PWD reconstructs content.

These sources strengthen a precise bridge: receiver state and phase can alter effective response, and distinct oscillatory regimes can correlate with different pathways. The PWD delta remains the joined proposition that an edge-conditioned departure vector adds reproducible information or causal leverage beyond those established component measures.

9.2 Evidence-to-claim matrix

Source result	Supported component	What remains unsupported
Huxter et al. [4]	rate and phase can carry dissociable variables	PWD baseline and spatial operator
Voloh et al. [3]	phase-of-firing information can exceed phase-blind rate information	full PWD vector or consciousness interpretation
Muller et al. [5] and Aggarwal et al. [6]	population activity can propagate as measurable waves	receiver-specific causal reconstruction

Source result	Supported component	What remains unsupported
Cardin et al. [13]	input timing relative to a driven gamma state can change evoked response	universal receiver window or complete PWD mechanism
Colgin et al. [14]	oscillatory regimes can distinguish pathway coupling in hippocampus	frequency-only routing law or general neural code

10. Open-data development benchmark

The primary test is a leakage-safe comparison on multichannel neural data with known or estimable geometry and one independently measured target. The model ladder is:

Model	Information admitted
M0	rate, event count, and evoked magnitude
M1	M0 plus amplitude or power
M2	M1 plus scalar phase locking or coherence
M3	M2 plus static structural or functional connectivity
M4	M3 plus leakage-safe recurrent history
M5-raw	flexible graph or sequence model with complete raw signals, geometry, and history
M6-PWD	same raw access and capacity as M5-raw plus the frozen PWD vector

The primary endpoint is

$$\Delta_{\mathrm{PWD}} = \mathcal{L}(M_{\text{best eligible baseline}}) - \mathcal{L}(M6_{\mathrm{PWD}}), \tag{19}$$

on a sealed test set, with loss, smallest effect of interest, uncertainty interval, multiplicity rule, grouping level, and replication requirement declared before opening the test outcome.

Equation (19) states the intended confirmatory endpoint for a fully eligible model ladder. The reported analysis is a development-stage approximation using a fixed random-feature Poisson family, not the strongest conceivable raw graph or sequence model and not the unopened five-session final test. The labels M5-raw and M6-PWD therefore describe the target comparison architecture; the actual development branches are the narrower, exactly capacity-matched implementations specified in Section 10.2.

For a receiver-phase intervention, define a separately powered interaction estimand

$$\Gamma_{\mathrm{phase}} = [Y(I, \phi_{\mathrm{pref}}) - Y(C, \phi_{\mathrm{pref}})] - [Y(I, \phi_{\mathrm{anti}}) - Y(C, \phi_{\mathrm{anti}})]\,, \tag{20}$$

where I and C are intervention and matched-control conditions and preferred versus anti-preferred phases are localized independently of the confirmatory endpoint. A nonzero Γ_{phase} is evidence for phase-specific modulation only when stimulation energy, rate, power, movement, arousal, and evoked artifacts are matched or modeled. It is not by itself evidence for PWD, NAPOT, content, or consciousness.

10.1 Adversarial decision table

Favorable-looking observation	Strong alternative	Required discriminator
PWD decoder beats rate and power	raw temporal model was too weak	equal-input M5-raw with matched capacity and tuning
phase term predicts behavior	waveform shape or evoked latency	cycle-shape controls, aperiodic controls, and latency-field null
spatial gradient predicts target	volume conduction or common input	source / lag controls, spatial nulls, and independent direction
preferred phase changes response	stimulation artifact or excitability drift	phase-matched sham, energy matching, and independent localizer
effect repeats within sessions	session leakage	subject / session grouped replication
high surprisal predicts report	salience or motor preparation	nuisance model and independently defined content endpoint

All preprocessing parameters that learn from data must be fit inside the training fold. A favorable decoder is not a causal result. Causal language requires a phase-specific intervention that changes transfer or the target in the receiver-specific direction predicted before intervention.

10.2 Frozen Allen Visual Coding Neuropixels execution profile

Draft 3 froze protocol PWD-D3-ALLEN-VC-NPX-V1 before neural-value access. The source is the Allen Institute Visual Coding Neuropixels DANDI release `000021`, immutable version `0.251116.2246`, DOI 10.48324/dandi.000021/0.251116.2246 [17,18]. The primary edge class is anatomically declared `LGd -> VISp`, and the target is each eligible receiver unit's spike count in the next 20 ms.

The metadata intake selected 21 sessions with exact-acronym LGd and VISp LFP coverage. Five were sealed as a final test and 16 were assigned to development. Before primary outcomes were computed, two development sessions were found to lack finite CCF coordinates for their selected VISp probe. Because the frozen 25-feature operator requires source-receiver distance, those sessions were excluded without coordinate imputation. Fourteen development sessions remained. This is a post-shakedown metadata-eligibility correction, not outcome-based trimming.

The predeclared shakedown session first exposed a comparator defect: adding PWD regenerated the entire random projection, so the raw subspace was not identical between models. That unfavorable v1 result remains preserved but is not used as the grouped estimate. Comparator v2 repaired the comparison, was locked before the remaining eligible development sessions were analyzed, and made no further model or endpoint changes after grouped execution began.

Comparator v2 used identical 16-feature shared raw projections in both branches. The raw-only branch received eight additional raw projections; the PWD branch received eight projections of 25 frozen PWD features. Both models had hidden width 24, 25 trainable output parameters, train-only standardization, ridge 3.0, and the same three projection seeds. Each session used its first contiguous 80% of eligible samples for fitting and final contiguous 20% for evaluation. Unit effects were medians across seeds, session effects were medians across eligible units, and the cross-session estimate was the median of 14 session effects.

10.3 Grouped development result

The grouped procedure analyzed 908,295 causal 20-ms samples and 921 eligible VISp units. The primary increment was capacity-matched raw-only Poisson deviance minus raw-plus-PWD Poisson deviance, so a positive value favored PWD. All 14 session medians were negative.

Grouped development summary	Value
Coordinate-eligible development sessions	14
Development sessions excluded for undefined CCF coordinates	2
Eligible VISp units across sessions	921
Causal 20-ms samples	908,295
Training / evaluation samples	726,635 / 181,660
Projection seeds per unit	3
Cross-session median raw-minus-PWD deviance	-0.001099
Cross-session mean raw-minus-PWD deviance	-0.001114
Session-bootstrap 95% interval for median	[-0.001301, -0.000872]
Sessions with positive primary increment	0/14
Two-sided exact sign-test p value	0.000122

The interval is a deterministic 20,000-resample session-level percentile bootstrap using seed `20260719`. The sign test omits exact zeros; there were none. Both summaries are descriptive development statistics, not confirmatory inference. The result fails the tested model's primary development criterion and favors the capacity-matched raw-only branch in this preparation.

10.4 Nulls and component ablations

Every required control was evaluated with the same shared raw branch and matched eight-feature capacity. For null comparisons, negative PWD-minus-null deviance favors intact PWD.

Comparison	Median across sessions	Bootstrap 95% interval	Direction across sessions	Interpretation
PWD minus within-block circular shift	-0.000362	[-0.000398, -0.000278]	14/14 negative	intact temporal alignment beat shifted PWD
PWD minus source-receiver role swap	-0.000158	[-0.000206, -0.000117]	14/14 negative	original declared direction beat role swap
PWD minus phase ablation	+0.000189	[+0.000136, +0.000210]	14/14 positive	removing phase terms improved prediction
PWD minus amplitude-duration ablation	-0.000154	[-0.000219, -0.000123]	14/14 negative	amplitude-duration terms helped relative to phase-only PWD

Each all-one-direction sign test has two-sided exact p=0.000122. These controls show structure inside the failed model, but they do not overturn the primary result. The strongest bounded reading is that temporal alignment, declared direction, amplitude departure, and event duration carried some information relative to their matched perturbations, while the configured phase-departure and lagged-phase-gradient terms degraded held-out prediction. The full PWD branch still performed worse than an equally sized additional-raw-feature branch.

10.5 Decision after development

The endpoint decision is now closed. The frozen 25-feature PWD model is not promoted as empirically supported, and its five-session final test will not be opened. Spending a scarce untouched holdout on a model that failed the capacity-matched primary comparison in 14 of 14 development sessions would add little discrimination while destroying the holdout's value for a scientifically motivated successor.

The original protocol therefore ends with the adverse development result reported here. The five session identifiers remain sealed. A successor may use them only if all of the following conditions are met before any outcome is accessed:

1. the successor has a new model and protocol identifier rather than reusing the failed protocol's identity;
2. its phase representation, baselines, edge rules, target, exclusions, model capacity, estimand, smallest effect of interest, multiplicity rule, and failure conditions are learned or justified using development data and independent literature only;
3. its code, dependency environment, seeds, and exact input manifest are frozen and hashed;
4. its primary comparator is at least as strong and equally informed as the successor branch;
5. the adverse 14-session result and the reason for redesign remain visible; and
6. the five-session outcome is executed once under the frozen successor with no adaptive repair.

Independent mathematical, signal-processing, and systems-neuroscience review remains invited and should inform the successor design, but it is not required to report this completed adverse result. No successor protocol is claimed to be frozen by this paper.

11. Falsification and narrowing rules

The current development result triggers the equal-input-raw failure condition for this 25-feature implementation in the bounded LGd-to-VISp prediction task. The broader operator family remains a research question, but this exact implementation cannot be represented as supported. More generally, the scoped empirical PWD claim is narrowed or rejected for a preparation when:

1. PWD features are unreliable under the preregistered estimator multiverse;
2. their held-out increment lies within the equivalence margin relative to the strongest eligible comparator;
3. apparent effects disappear under power-, waveform-, reference-, or leakage-preserving controls;
4. direction or delay cannot be independently constrained and the equivalence class is too broad;
5. spatial patterns are explained by common input, evoked latency, volume conduction, or movement;
6. a phase-specific intervention fails to alter receiver transfer despite adequate manipulation;
7. an equal-input raw model reproduces the endpoint without the claimed PWD-specific structure; or
8. the result fails across the preregistered replication units.

Failure of this operator would not imply that neural timing or traveling waves are unimportant. It would show that this receiver-relative decomposition did not add the proposed value in the bounded test.

12. Relationship to SAN and NAPOT

The 2017 ATP proposal asked which feedback and broadcast modes a biological neural network uses; PWD later proposed one receiver-relative measurement vocabulary for changes carried through such a network. ATP is therefore a protocol-search ancestor, not an empirical validation of PWD. Within SAN, the tonic state is a maintained context or "canvas" and a phasic differential is an update or "ink." The mathematical translation is a past-only baseline and a candidate departure. The network itself, through distributed receiving, state change, projection, action, and feedback, is the observer-action process; PWD does not introduce a hidden viewer.

NAPOT uses PWDs as possible inputs to a larger recurrent reconstruction model [2]. PWD is therefore neither equivalent to NAPOT nor sufficient for it. A PWD paper asks whether the differential is well-defined, identifiable, informative, and causally useful. A NAPOT paper asks whether multiple distributed projections constrain an independently testable latent state and its action-feedback consequences. Keeping the levels separate lowers the claim burden of both papers.

13. Limitations

This Draft 6 is a formalization, software-contract report, and grouped development analysis. It is not a confirmatory result. The comparator is one fixed random-feature Poisson model family, not the strongest conceivable raw sequence or graph model. The edge is one anatomically declared LGd-VISp channel pair per session. The target is short-horizon spike count during natural scenes, not perceptual content, report, action, or consciousness. Two development sessions were ineligible because receiver coordinates required by the spatial operator were undefined.

The complex filter is one estimator among many. Event thresholds and baseline constants are not derived from cellular physiology. The vector mixes circular, continuous, integer, and spatial quantities. The positive relative controls could reflect generic nonlinear temporal structure rather than the intended PWD interpretation. The phase-ablation advantage could arise from phase-estimation noise, an unsuitable filter or lag, the spatial-gradient definition, projection inefficiency, or a genuine absence of useful phase information for this target. Development data cannot adjudicate among these explanations without a new, prospectively frozen discrimination program.

The paper does not show that surprisal is meaning, information is consciousness, or a phase-coded artificial system is conscious. It also does not show that all receiver-relative neural differences are useless. It shows that this complete 25-feature implementation failed to add held-out predictive value over the matched raw-capacity comparator in these 14 development sessions.

14. Conclusion

A phase-wave differential can be stated without metaphor as a receiver-relative vector of phase, amplitude, duration, and spatial-lag departures from a strictly past baseline. The operator has an executable causal implementation and explicit failure conditions. Existing neural studies support component ideas, but they do not validate the complete operator.

The grouped development result is adverse and decisive for the tested implementation: all 14 session-level primary effects favored the capacity-matched raw-only branch. Required controls also showed that intact timing and declared direction beat their perturbations, amplitude-duration terms helped relative to phase-only PWD, and the configured phase terms hurt. This pattern is scientifically more useful than a binary slogan. It rejects promotion of the full current model, identifies a specific phase-representation problem, and preserves a reproducible basis for a successor test.

The sealed final sessions remain unopened. The recorded decision is to preserve them rather than test the failed development model. The next legitimate empirical step is a separately named, prospectively frozen successor learned only from development evidence and executed once on the untouched holdout. No conclusion about NAPOT, Neural Rendering, qualia, or consciousness follows from this benchmark.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research presented in this paper. The underlying research program was not written or developed by AI. This body of work is Micah Blumberg's original research, developed through a continuous program spanning 2011-2026. Its central conceptual foundations were developed and recorded before ChatGPT's public release on November 30, 2022.

In later manuscript development, AI tools were used only in bounded supporting roles, including assistance with mathematical and computational formulation, code, grammar and spelling, and checks against medical, biological, and physics literature. These tools did not supply the original theories or authorship. Micah Blumberg selected the claims and sources, directed the analysis, approved the final language, and remains responsible for the paper's arguments, interpretations, and errors.

Data and code availability

No repository release or public release is authorized by this local package. Draft 6 analyzes 14 development sessions from the previously published Allen Institute mouse Visual Coding Neuropixels dataset, DANDI `000021`, immutable version `0.251116.2246`. No new animal experiment was performed. Two development sessions were excluded because finite receiver CCF coordinates required by the frozen spatial feature were unavailable; no coordinates were imputed and no primary outcomes were computed for those sessions.

The frozen protocol, comparator lock, execution supplement, coordinate-eligibility manifest, 14 per-session result files and validation receipts, deterministic grouped aggregate, and aggregate validation receipt are stored under `protocols/`, `manifests/`, `results/`, `scripts/`, and `validation/` in this governed package. The five-session final test remains unopened. This final-preprint successor reports the adverse development endpoint; it does not freeze or execute a successor protocol.

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